

Python and Java Code Notes

1. Python Basics

```
# Python: simple program
name = "Abhishek"
age = 20

def greet(name):
    return f"Hello, {name}!"

print(greet(name))
print("Age:", age)
```

2. Python Function

```
# Python: loop and list
numbers = [1, 2, 3, 4, 5]

for number in numbers:
    print(number * 2)
```

3. Java Basics

```
// Java: simple program
public class Main {
    public static void main(String[] args) {
        String name = "Abhishek";
        int age = 20;

        System.out.println("Hello, " + name);
        System.out.println("Age: " + age);
    }
}
```

4. Java Class and Method

```
// Java: class and method
class Calculator {
    int add(int a, int b) {
        return a + b;
    }
}

public class Main {
    public static void main(String[] args) {
        Calculator calc = new Calculator();
        System.out.println(calc.add(10, 20));
    }
}
```

5. Comparison Table

Feature	Python	Java
Typing	Dynamically typed	Statically typed
Syntax	Short and concise	More verbose

Execution	Interpreted / bytecode	Compiled to bytecode
Common Use	AI, data science, automation	Backend, enterprise, Android